

Nearby AirSend

Product Requirements Document (PRD) + Developer Logic Flow

Feature Name: Nearby AirSend

Goal:

Enable fast nearby peer-to-peer transfers using gesture, BLE, NFC, and UWB proximity technologies.

User Flow:

Sender:

Tap Local Send → Search nearby devices → NFC handshake / BLE scan → Connect recipient

→ Enter amount → Authenticate PIN/biometric → Pick funds → Air drop → Confirm → Success

Recipient:

App open → Connected by sender → Waiting for drop → Accept transfer → Wallet credited

Technologies:

1. NFC for instant tap connection

- iPhone: top-to-top
- Android: back-to-back

2. BLE for persistent nearby session

- connection sync
- device identity
- transfer token

3. UWB for directional precision and fast targeting

Security:

- PIN / biometrics
- encrypted session token
- recipient preview
- transaction confirmation
- timeout
- auto disconnect

Developer Logic:

Sender:

onNfcConnect() -> createSession() -> switchToBLE()

onPickGesture() -> walletState = "picked"

onDropGesture() -> sendPendingTransfer()

onRecipientAccept() -> processTransfer()

Backend Flow:

Create pending transaction

sender_id

receiver_id

amount

session_token

status = pending

After recipient accepts:

debit sender

credit recipient

status = completed

push notification to both devices